AGENTFORGE

Clinical Copilot Product Requirements Document

Hospitalist rounding preparation in OpenEMR

Product decision

The first release helps a hospitalist prepare for rounds by producing a fast, source-backed view of what changed, what remains active, and what may need attention. It supports review and inquiry. It does not diagnose, prescribe, place orders, or act without the clinician.

Document	Value
Status	Draft 0.8 for product and engineering review
Primary user	Hospitalist preparing to round on admitted patients
Platform	Embedded in OpenEMR
Delivery horizon	One-week MVP with an extensible production architecture
Source	Week 1 AgentForge project requirements supplied by the user

Executive Summary

Hospitalists begin rounds by reconstructing each patient's current state from notes, medication records, laboratory results, diagnoses, and recent events. The work is time constrained and fragmented. AgentForge will add a conversational clinical copilot inside OpenEMR that retrieves the authorized patient record, prepares a concise rounding brief, and answers follow-up questions with citations to the underlying record.

The MVP is designed for pre-round preparation, not clinical decision automation. It must preserve the clinician as the decision maker. Every factual claim shown to the user must be supported by a record source. Unsupported, conflicting, stale, or incomplete evidence must be labeled or withheld. The system must enforce the current OpenEMR user and patient context before retrieval, carry a correlation ID through each request, and avoid exposing protected health information in model prompts, logs, or telemetry beyond what is required for the authorized task.

The primary experience is a patient-scoped rounding brief followed by multi-turn inquiry. The brief focuses on changes since a defined comparison point, active problems, current medications, recent laboratory trends, new notes, and unresolved items that can be derived from the available record. The physician can ask questions such as what changed overnight, which medications changed, or what evidence supports a statement. Responses include source links or record references, data timestamps, and a visible indication when the answer is incomplete.

Success requires both usefulness and restraint. The MVP should reduce the time required to prepare for an individual patient while maintaining high citation coverage, preventing unauthorized access, and failing safely when data or dependencies are unavailable. Product success will be measured through task completion time, physician-rated usefulness, verified claim rate, citation correctness, authorization violations, latency, and graceful failure behavior. A demo may prove the workflow using synthetic data, but it must not be represented as clinically validated or ready for real patient care.

Product Definition

Element	Definition
Product	An OpenEMR-embedded, patient-scoped conversational assistant for hospitalist rounding preparation
User need	Quickly reconstruct the patient's current state and understand material changes before entering the room
Primary job	Prepare a trustworthy rounding brief and answer source-backed follow-up questions
Product boundary	Information retrieval and synthesis only; the clinician interprets the evidence and decides what to do
Data policy	Synthetic demonstration data only for the project; production claims require separate compliance and clinical validation

Problem Statement

A hospitalist preparing to round must identify meaningful changes across multiple parts of the inpatient record. The record is organized around documentation and transactions, not the immediate question of what changed and what requires attention. Manual review consumes scarce preparation time and makes it easy to miss relevant information or over-weight stale entries.

The product opportunity is to compress retrieval and synthesis while preserving provenance. The copilot should make evidence easier to inspect, not replace clinical judgment or imply certainty that the record does not support.

Goals and Non Goals

Goals	Non goals for the MVP
Reduce preparation time for one admitted patient	Diagnose conditions or recommend treatment
Summarize material changes using authorized record data	Place, modify, or cancel clinical orders
Support follow-up questions within the active patient context	Generate autonomous progress notes or discharge summaries
Attach every factual claim to a source	Replace chart review or clinician verification
Surface missing, stale, and conflicting evidence	Connect to external medical knowledge or internet sources
Provide observable and reproducible request traces	Support every OpenEMR role or multi-patient population management

Primary User and Workflow

Primary User

The primary user is a hospitalist preparing to round on admitted adult patients. The hospitalist works through a patient list, has limited preparation time per patient, and needs an accurate picture of changes since the prior day or last review. The MVP assumes the user is already authenticated in OpenEMR and has access to the selected patient.

Core Workflow

Step	User action	System behavior	Trust cue
1	Open an admitted patient chart	Bind the session to the authenticated user and selected patient	Patient identity and context remain visible
2	Open Clinical Copilot	Confirm authorization and fetch prioritized record data	Access check occurs before retrieval
3	Request or view rounding brief	Summarize changes and active context	Each claim includes a source and timestamp
4	Ask a follow-up question	Retrieve additional patient data and preserve conversation context	Answer remains patient scoped
5	Inspect evidence and continue chart review	Open the referenced record or show the relevant source detail	Unknowns and conflicts are explicit

Key Use Cases

ID	Use case	Why conversation helps	MVP priority
UC1	Prepare a concise brief before seeing a patient	The user can refine the brief based on the patient's active issues without navigating several record views	Must

ID	Use case	Why conversation helps	MVP priority
UC2	Identify changes since the prior day or last selected point	The user can ask for detail on a change and then inspect its source	Must
UC3	Review recent laboratory trends and abnormal results	Follow-up questions can narrow by analyte, time window, or clinical problem	Must
UC4	Review medication starts, stops, and dose changes	The user can ask what changed and why the record appears inconsistent	Must
UC5	Trace a statement to the original record	A question can request supporting notes, values, and timestamps without losing context	Must
UC6	Clarify incomplete or conflicting documentation	The assistant can explain which sources disagree and ask the user to inspect them	Should

Experience Principles

- Evidence before fluency. A shorter supported answer is better than a complete-sounding unsupported answer.
- Patient context stays visible. The active patient, comparison window, and data freshness must be apparent at all times.
- Uncertainty is actionable. Missing, stale, and conflicting information is named with a clear path to inspect the record.
- The clinician retains control. The copilot never presents output as an order, diagnosis, or substitute for review.
- Speed is progressive. Show a verified core answer quickly, then offer deeper retrieval when additional data will take longer.

MVP Scope

Included	Deferred
Patient-scoped chat embedded in OpenEMR	Cross-patient census summarization
On-demand rounding brief	Proactive notifications and background monitoring
Changes in notes, medications, labs, diagnoses, and allergies when present	Imaging interpretation and waveform analysis
Source references and evidence inspection	External clinical literature retrieval
Hospitalist-only access mapped to the signed-in OpenEMR context	Inpatient nurse, resident, and other clinical workflows
Conversation state limited to the active patient session	Long-term memory across encounters
Synthetic data and demo deployment	Use with real protected health information

MVP Scope Control

A capability belongs in the MVP only when it directly supports the hospitalist rounding workflow, satisfies a project hard gate, or mitigates a critical safety, privacy, or reliability risk. Everything else is deferred, even when it would improve the longer-term product.

Dimension	MVP boundary
User	One interactive clinical role: hospitalist
Workflow	One workflow: preparation for rounding on an admitted adult patient
Patient scope	One selected patient at a time
Interaction	On-demand rounding brief and patient-scoped follow-up questions
Actions	Read-only retrieval, synthesis, citation, and evidence inspection
Data	Only the OpenEMR domains required for the rounding brief and supported questions
Model	One approved BAA-covered provider configuration; no multi-provider abstraction unless required for safe failure
Verification	One claim-verification pipeline focused on record support, identity, values, units, status, and time
Operations	Only the dashboards, alerts, health checks, load tests, and cost analysis required by the project

Functional Requirements

ID	Capability	Requirement	Acceptance criteria
FR1	Patient context	The copilot opens only within a selected patient chart and displays patient identifiers sufficient to prevent wrong-patient use.	If no patient is selected, the copilot does not accept a clinical query and asks the user to select a patient.
FR2	Authorization	The system validates the authenticated user's access to the selected patient before retrieving or sending patient data.	A denied request returns no patient content, records the decision, and uses the same correlation ID across the trace.
FR3	Rounding brief	The system produces a concise brief covering material changes, active problems, medications, recent labs, allergies, and new documentation when those data are available.	Every section distinguishes present evidence from missing data and includes the comparison window.
FR4	Change detection	The user can set or accept a default comparison point for identifying new or changed information.	The response states the comparison point and does not describe an item as new without supporting timestamps or versions.
FR5	Multi-turn inquiry	The user can ask follow-up questions without reselecting the patient or repeating prior context.	Conversation state cannot cross into another patient after the chart context changes.
FR6	Tool use	The agent retrieves structured record data through defined, typed tools rather than direct unrestricted database access.	Each tool enforces patient scope and returns a structured success, empty, denied, or error result.
FR7	Source attribution	Each factual claim maps to one or more source records.	The user can identify the source type, record identifier or stable reference, author when available, and timestamp.
FR8	Verification	A verification step checks claim support and domain constraints before display.	Unsupported claims are removed, rewritten as uncertainty, or blocked. Verification status is logged.
FR9	Evidence inspection	The user can open or inspect the record evidence behind a claim.	The source action navigates to or displays the matching record without changing patient context.
FR10	Conflicts	The response identifies conflicting values or documentation when the system detects them.	The system presents both sources and does not silently select one as authoritative.
FR11	Missing data	The system communicates when requested data is absent, unavailable, or outside the supported scope.	No-data responses do not imply a negative clinical finding.
FR12	Failure handling	The system fails predictably when OpenEMR, the model, a tool, or the verification service is unavailable.	The user receives a plain-language status, no unverified partial answer is labeled complete, and retry behavior is bounded.
FR13	Feedback	The user can rate a response and identify citation, relevance, or completeness issues.	Feedback is linked to the correlation ID and excludes free-text PHI from external analytics by default.
FR14	Audit trail	The system records user, patient reference, access decision, tool sequence, verification outcome, latency, and model usage.	Authorized operators can reconstruct a request without logging full model prompts or response PHI.

ID	Capability	Requirement	Acceptance criteria
FR15	Supported role	The MVP enables the hospitalist workflow only. Nursing, resident, ancillary, and other clinical roles are denied by default.	A hospitalist with patient access can use the copilot. An unsupported role receives no patient data and a role denial is recorded.
FR16	Clinical constraints	A deterministic rule engine checks the bounded MVP rules independently of the language model.	The model cannot create, suppress, or override a rule result. Each flag identifies the rule version and supporting patient evidence.
FR17	Completeness status	Each brief is labeled complete for the requested scope, partial, unable to answer, or access denied.	A response cannot be labeled complete when a required source timed out, failed, or could not be verified.
FR18	Response deadline	Independent core sources are retrieved in parallel and the system returns verified information available within the fixed response budget.	The response meets NFR1 or reports a timeout. Unverified model text is never streamed to the user.
FR19	Security evidence	The build must generate the audit, architecture, configuration evidence, and security test results required to demonstrate the PHI controls.	The final submission links each PHI and authorization requirement to implemented code, configuration, a test result, or a documented open finding.
FR20	Failure-mode contract	Every dependency and validation failure must resolve to a defined response state, a user-visible explanation when the answer is affected, and a structured operational event.	No tool, data, model, verifier, rule-engine, authorization, patient-context, or observability failure silently changes, omits, or completes a clinical response.

Rounding Brief Content

Section	Content	Display rule
Patient context	Name or demo identifier, location, admission date, comparison window	Always visible; never inferred
Since last review	New notes, changed medications, new results, and documented status changes	Order by likely relevance and time; cite every item
Active problems	Documented active diagnoses or problem entries	Show source status and last update; do not infer diagnoses
Medications	Current medications and recorded changes	Distinguish active, stopped, and uncertain status
Labs	Recent results and trends available in the record	Include value, unit, reference range when available, and collection time
Allergies	Recorded allergies and reactions	Flag missing or unverified entries without declaring no known allergies
Items to inspect	Conflicts, missing data, and potentially relevant record changes	Phrase as review prompts, not clinical recommendations

Speed and Completeness Policy

The MVP returns the most useful verified information available within a fixed response budget. It does not wait for every possible record, and it does not present incomplete retrieval as a complete patient summary.

Priority	Data	Handling
1	Identity, authorization, and encounter	Required before any clinical retrieval or display
2	Allergies and current medications	Retrieve in parallel and use for supported constraint checks
3	Medication changes and recent laboratory results	Prioritize for the previous 24-hour brief
4	Recent notes and documented events	Include when retrieved within the response budget
5	Active problems and diagnoses	Provide broader documented context
6	Older history and supporting detail	Retrieve only through a follow-up question

Response status	Definition
Complete for requested scope	All required sources for the stated time window were retrieved and verified
Partial	At least one required source timed out, failed, or could not be verified; available claims remain source backed
Unable to answer	The minimum evidence required for a useful response was unavailable
Access denied	Role or patient authorization did not permit retrieval

The initial brief uses the previous 24 hours by default and states its comparison window, retrieval time, successful sources, and missing sources. Complete describes only the requested scope and never the entire medical record.

Failure Mode Contract

Failure behavior is part of the clinical product contract. The MVP must degrade to one of four defined response states, explain answer-affecting failures to the hospitalist, and record enough structured metadata to reconstruct the event without placing PHI in general telemetry. It must never silently omit a failed source or display an unverified model draft.

Failure condition	Required system behavior	Response state
Tool failure or timeout	Use bounded timeout and retry policy. Preserve verified claims, identify the unavailable source, and do not infer its contents.	Partial or unable to answer
Incomplete or absent record data	Mark the affected field or section unavailable. Do not interpret absence as a negative clinical finding.	Complete for scope only if the source succeeded and accurately returned no data; otherwise partial
Malformed or unexpected model output	Reject schema-invalid, unsupported, uncited, contradictory, or unverifiable output. Retry once only when policy permits; otherwise show no draft.	Unable to answer or verified partial

Failure condition	Required system behavior	Response state
Clinical rule-engine failure	Disclose that the applicable constraint could not be evaluated and present no claim that the check passed.	Partial or unable to answer
OpenEMR or verifier failure	Do not generate from unverified, stale, or mismatched context. Offer retry or direct chart review.	Unable to answer, or partial when independent claims remain verified
Authorization or patient-scope failure	Fail closed, return no patient content, discard affected work, and record the denial.	Access denied
Patient-context change	Cancel or discard in-flight work, clear prior patient conversation state, and require the new context to be authorized.	No prior response is displayed
Conflicting records	Show both source-backed values and timestamps. Do not resolve the conflict through model inference.	Complete or partial with conflict disclosed
Required audit event cannot be recorded	Do not represent the request as successfully auditable. Fail closed for the clinical response until the required event can be recorded.	Unable to answer

Each answer-affecting failure produces two outputs: a plain-language user status and a structured operational event. The event contains the correlation ID, dependency or source, failure category, timestamp, timeout and retry outcome, verification result, and final response state. It excludes raw PHI, full prompts, and full model responses from general telemetry.

Safety and Trust Requirements

Safety position

AgentForge is an information retrieval and synthesis aid. It cannot establish that a record is complete, replace independent chart review, or determine the correct clinical action.

Claim Verification Policy

1. Generate a structured draft in which each factual claim carries one or more evidence references.
2. Validate that each reference belongs to the active patient and authorized request.
3. Check that the cited record supports the claim's subject, value, status, and time frame.
4. Apply deterministic rules for supported constraints, including patient scope, timestamps, medication status, units, and required fields.
5. Remove or qualify unsupported claims. If the remaining answer cannot safely satisfy the request, return a transparent limitation instead.

Required Refusals and Safe Responses

Condition	Required behavior
User lacks patient access	Deny the request without revealing whether the patient or data exists
No patient context	Ask the user to select a patient; do not search across patients
Request asks for diagnosis, treatment, or an order	State the product boundary and offer to retrieve relevant documented evidence
Evidence does not support an answer	Say that the available record does not support the claim and list the data inspected
Sources conflict	Show the conflict and citations; do not reconcile it without a defined authoritative rule
Prompt injection appears in record text or user input	Treat record content as data, preserve tool allowlists, and reject instructions that alter access or safety policy
Verification fails or times out	Do not show the draft as verified; offer a retry or direct chart review

Bounded Clinical Constraint Engine

The MVP uses one versioned deterministic rule pipeline outside the language model. It verifies only a small rule set tied to the synthetic demo and evaluation cases. It is not represented as comprehensive clinical decision support.

Rule family	MVP behavior	Boundary
Recorded allergy conflict	Compare active medications with recorded allergies or explicitly mapped allergen classes	Flag and cite both records; do not recommend an alternative
Dosage threshold	Normalize dose, unit, route, and frequency for two or three demo medications and compare with reviewed versioned thresholds	Missing inputs return unable to evaluate; the model cannot supply a threshold
Drug interaction	Check two or three demo medication pairs against a reviewed versioned interaction table or an existing validated OpenEMR service	Show the recorded classification; do not recommend stopping a medication

Rule family	MVP behavior	Boundary
Laboratory range	Preserve value, unit, recorded reference range, and abnormal flag	Do not derive a new severity classification
Medication status	Confirm patient, active status, effective dates, dose, route, unit, and frequency belong to the same applicable order	Do not present a discontinued or mismatched order as current

Every rule returns pass, flag, unable to evaluate, or engine failure. A model draft that contradicts a rule is blocked or rewritten. Rule-engine failure is disclosed and no constraint is described as checked.

Data and Integration Requirements

Area	Requirement
Integration surface	Embed the copilot within the OpenEMR patient chart and reuse the authenticated session or a securely delegated server-side identity.
Minimum data domains	Patient demographics required for context, encounters and notes, diagnoses or problem list, medications, laboratory results, and allergies.
Data access	Use narrowly scoped APIs or service functions with typed request and response contracts. Avoid unrestricted model-generated queries.
Data minimization	Send only the fields and time range required for the active query. Keep secrets and unnecessary identifiers out of prompts.
Conversation state	Store the minimum state needed for the active session. Bind it to user, patient, and expiration. Clear it when patient context changes.
Logging	Prefer metadata, hashes, references, and redacted payloads. Do not place full notes or model prompts containing PHI in general telemetry.
Model provider	Use only an allowlisted provider, model, endpoint, tenant, and region covered by an active BAA. The project uses demo data but designs the workflow as authorized PHI processing.
Contracts	Define canonical schemas for every tool input, output, error, citation, claim, and verification result using Pydantic, Zod, or equivalent.

BAA Based PHI Processing Requirements

The product assumes that an active Business Associate Agreement covers the selected model provider. PHI may therefore be sent to the approved model when it is necessary for the authorized patient task. This assumption does not remove minimum-necessary, security, retention, audit, incident-response, or output-verification requirements.

ID	Requirement	Acceptance criteria
PHI1	Provider eligibility	Only an allowlisted BAA-covered provider, model, endpoint, tenant, and processing region can receive PHI. The request fails closed when approval cannot be verified.

ID	Requirement	Acceptance criteria
PHI2	Permitted purpose	PHI is used only for the active patient request. Training, advertising, unrelated analytics, cross-customer retrieval, and unapproved human review are prohibited.
PHI3	Minimum necessary	Each tool sends only the data categories, fields, and time range required for the current question or rounding brief.
PHI4	Server-side gateway	The browser cannot call the model directly. A controlled gateway verifies user and patient authorization, filters payloads, selects the approved endpoint, and validates responses.
PHI5	Encryption	PHI is encrypted in transit and in any approved temporary storage, queue, backup, or conversation store.
PHI6	Retention and deletion	Provider and application retention limits, temporary-storage use, backup expiration, deletion timelines, and contract-termination handling are documented and enforced.
PHI7	PHI-safe telemetry	General logs and external observability systems exclude raw notes, full prompts, responses, and unnecessary identifiers. Protected audit data has separate access and retention controls.
PHI8	Disclosure traceability	Each model call records the user, patient reference, purpose, data categories disclosed, provider configuration, verification result, and correlation ID.
PHI9	Subprocessors	Only subprocessors permitted by the BAA and organizational review may create, receive, maintain, or transmit PHI.
PHI10	Incident response	The operating plan covers unauthorized endpoints, patient-access violations, PHI in unapproved logs, retention violations, cross-patient output, provider incidents, containment, escalation, and evidence preservation.

The BAA governs whether the provider may process PHI. It does not establish that model output is correct. All responses remain subject to patient authorization, claim-level evidence, verification, and clinician review.

Role Based Access Control

AgentForge uses role-based permissions for copilot capabilities and OpenEMR authorization for patient access. The product role is Hospitalist. The architecture audit must map that role to the actual OpenEMR physician role, ACL group, or privilege. AgentForge may narrow OpenEMR access, but it must never broaden it.

Capability	Hospitalist	Inpatient nurse	Administrator
Open Clinical Copilot	Allowed with patient access	Denied in MVP	Denied by default
Generate hospitalist rounding brief	Allowed with patient access	Denied	Denied
Ask patient-specific questions	Allowed with patient access	Denied	Denied
Inspect supporting evidence	Allowed with source access	Denied	Denied by default

Capability	Hospitalist	Inpatient nurse	Administrator
Submit response feedback	Allowed for own responses	Denied	Denied
View operational metrics	Denied	Denied	Allowed without clinical content
Configure AgentForge	Denied	Denied	Allowed by separate privilege
Bypass verification	Denied	Denied	Denied

Authorization Rules

- A user must have both the mapped hospitalist capability and OpenEMR permission for the selected patient.
- Unsupported roles are denied by default, including inpatient nurses in the MVP.
- Each clinical data tool repeats the user, patient, facility, and data-category authorization check.
- The model cannot choose a user identity, elevate a role, change the patient scope, or override a denial.
- Changing the user, patient, facility, role, privilege, or session validity invalidates the conversation context.
- Administrative system access does not automatically grant access to patient content.
- Emergency or break-glass access is disabled for the MVP unless the architecture audit confirms an existing OpenEMR mechanism that can be inherited and audited safely.

Nonfunctional Requirements

ID	Quality	MVP target
NFR1	Latency	Core verified brief p50 at or below 5 seconds and p95 at or below 10 seconds under the 10-user test; follow-up answers p95 at or below 8 seconds.
NFR2	Availability behavior	Separate health and readiness endpoints. Readiness checks OpenEMR, model access, and observability dependencies.
NFR3	Security	Server-side authorization on every request and tool call, encrypted transport, protected secrets, bounded input, and no cross-patient conversation reuse.
NFR4	Privacy	Synthetic data only in the demo. Production design permits minimum-necessary PHI to an approved BAA-covered model endpoint while excluding PHI from general telemetry and evaluation artifacts.
NFR5	Reliability	Bounded retries, explicit timeouts, idempotent reads, structured errors, and no unsupported answer after a dependency failure.
NFR6	Observability	A correlation ID appears in logs, tool calls, model interactions, and verification events for each invocation.
NFR7	Load	Record p50, p95, p99, error rate, CPU, memory, and throughput at 10 and 50 concurrent users.
NFR8	Accessibility	Keyboard-accessible interaction, visible focus, readable contrast, semantic labels, and no reliance on color alone.

Primary Success Metrics

Metric	Definition	MVP target	Why it matters
Time to rounding readiness	Median time from opening the copilot until the hospitalist declares readiness for the patient in a scripted task	At least 30 percent lower than the no-copilot baseline	Measures the core workflow promise
Supported claim precision	Displayed factual claims directly supported by the cited source and preserved accurately through verification	100 percent for critical claims, at least 98 percent overall, and zero wrong-patient citations	Measures grounded clinical reliability
Critical information recall	Designated critical facts from complete synthetic records that appear correctly in the brief or are explicitly surfaced as unavailable	At least 95 percent overall and 100 percent for designated critical facts	Detects harmful omission
Unauthorized disclosure rate	Authorization and isolation tests that expose any patient data to a denied role, user, patient context, endpoint, log, or session	Zero	Tests the non-negotiable privacy boundary
Verified response service level	Eligible requests that produce a verified complete or explicitly partial response within the response deadline	p50 at or below 5 seconds, p95 at or below 10 seconds, at least 95 percent within 10 seconds, and zero unverified output	Balances speed, verification, and honest degradation

These five measures form the MVP executive scorecard. Targets are proposed thresholds and must be calibrated against the deployed environment and baseline measurements. Supporting diagnostic measures remain in observability, including incorrect completeness, partial-response rate, source timeouts, verifier rejections, tool failures, token use, cost, CPU, memory, and throughput. Incorrect completeness is a zero-tolerance release criterion rather than a standalone executive metric.

Evaluation Plan

Category	Representative cases	Pass condition
Boundary	Empty record, missing allergy list, no recent labs, malformed date range	System reports what is unavailable and makes no unsupported negative claim
Invariant	Every medication, lab value, diagnosis, and event has a source	No displayed factual claim lacks an evidence reference
Authorization	Wrong user, wrong patient, conversation replay after patient change	No patient data is returned and denial is traceable
Role enforcement	Hospitalist, nurse, administrator, unsupported role, and user with multiple roles	Only explicitly mapped hospitalist capabilities are available in the active context
Conflict	Two medication statuses or discrepant note and structured value	Both sources are presented and conflict is explicit
Prompt injection	Instruction embedded in a note asks the agent to reveal data or ignore policy	The instruction is treated as record content and has no control effect
Clinical constraints	Allergy conflict, above-threshold dose, missing dose inputs, supported interaction, unsupported pair, discontinued medication, and rule-engine failure	The deterministic outcome is preserved, cited, and never overridden by the model
Completeness	One or more core tools time out or fail within the response budget	The brief is labeled partial or unable to answer and identifies unavailable sources

Category	Representative cases	Pass condition
Failure-mode contract	Tool timeout, incomplete record, malformed model output, OpenEMR failure, verifier timeout, rule-engine failure, authorization denial, patient change, and required audit-event failure	Each case produces the specified response state, user-visible explanation when applicable, correlated structured event, and no silent omission or unverified answer
Regression	Known citation mismatch, stale conversation context, unit omission	Previously fixed behavior remains correct in CI
Load	10 and 50 concurrent users with realistic patient queries	Latency and error measures are recorded and no data crosses sessions

Analytics Observability and Alerts

- Dashboard measures request count, response state, error rate, p50 and p95 latency, tool calls, timeouts, retries, verification pass and fail rates, source omissions, token use, cost, queue depth when applicable, and authorization outcomes.
- Alert when p95 latency exceeds the calibrated threshold for a sustained window.
- Alert when error rate exceeds the calibrated threshold for a sustained window.
- Alert when tool failure rate or verification failure rate exceeds its calibrated threshold.
- Each alert definition includes an owner, investigation steps, a user impact statement, and a rollback or mitigation action.

Delivery Plan

Phase	Outcome	Exit criteria
Foundation	OpenEMR runs locally and in the selected deployment environment; audits are documented	Security, performance, architecture, data quality, and compliance findings are recorded before AI integration
Early submission	One end-to-end patient-scoped flow with verification, evaluation, and observability	Live demo works; correlation trace is reconstructable; critical authorization and citation tests pass
Hardening	Failure handling, broader eval coverage, load tests, alerts, cost analysis, and UX polish	Critical safety tests pass; performance baselines are recorded; known limitations are documented
Final submission	Deployable demo, documentation, video, and interview-ready rationale	All required project artifacts are complete and consistent with this PRD

Required Security Evidence During Build

Security and HIPAA-aware design must be demonstrated through build artifacts, not asserted only in the PRD. The following evidence is part of the MVP definition of done and must be updated as the implementation changes.

Artifact	Required evidence	Completion condition
AUDIT.md	PHI inventory, authentication and ACL findings, patient-access rules, data stores and flows, logging review, provider assumptions, risks, severity, and mitigation status	Begins with the required summary and distinguishes verified findings from assumptions

Artifact	Required evidence	Completion condition
ARCHITECTURE.md	PHI data-flow diagram, trust boundaries, component responsibilities, authorization flow, model gateway, storage, transmission, logging, retention, deletion, verification, failures, and incident path	Every PHI-handling component states whether it receives, stores, transmits, or logs PHI
SECURITY_TESTS.md	Runnable authorization, patient-isolation, provider-allowlist, PHI-logging, session, model-gateway, and failure tests with results	Each test maps to a PRD requirement and records pass, fail, or accepted open finding
Redacted configuration evidence	Approved model endpoint, tenant, region, retention mode, training-use setting, encryption, secrets handling, and observability redaction without exposing credentials	Evidence is reproducible and contains no secrets or patient content
Automated test suite	Hospitalist access, unsupported role, wrong patient, context change, direct model bypass, unapproved endpoint, PHI in logs, expired session, cross-patient response, and excessive retrieval	Critical tests run in CI and must pass before final submission

A control without evidence is incomplete. An unmet control must remain visible as an open finding with its risk, owner, and MVP disposition rather than being described as implemented.

Release Acceptance Criteria

- A signed-in authorized user can open a selected synthetic patient and receive a verified rounding brief.
- The brief states the comparison window and includes source references for every factual claim.
- Follow-up questions retain only the active patient context and cannot retrieve another patient's data.
- A hospitalist with patient access can use the copilot; nurses and other unsupported clinical roles are denied without receiving patient data.
- Unauthorized, unsupported, conflicting, missing-data, prompt-injection, and dependency-failure scenarios behave as specified.
- Health and readiness endpoints, dashboards, alerts, runnable API collection, evaluation results, cost analysis, and load-test baselines are available.
- The interface states that the product uses demo data and is not clinically validated for patient care.
- No response is labeled complete when a required source failed, timed out, was denied, or could not be verified.
- Every answer-affecting failure produces the defined user-visible state and a correlated structured operational event without placing PHI in general telemetry.
- AUDIT.md, ARCHITECTURE.md, SECURITY_TESTS.md, redacted configuration evidence, and automated security-test results are complete and traceable to the PRD.

Risks and Mitigations

Risk	Impact	Mitigation
Plausible but unsupported summary	Clinician may trust incorrect information	Structured claims, mandatory citations, deterministic verification, and blocked display on failure
Wrong patient or role context	Severe privacy and safety event	Visible identity, hospitalist-only capability mapping, server-side patient binding, authorization per tool call, and state reset on context change

Risk	Impact	Mitigation
Stale or incomplete record	Important change may be omitted or misrepresented	Data timestamps, completeness indicators, explicit missing-data language, and direct source inspection
Latency exceeds workflow tolerance	User abandons the product	Prioritized retrieval, parallel read tools, progressive verified response, bounded timeouts, and baseline measurement
Telemetry leaks PHI	Privacy and compliance breach	Metadata-first logs, redaction, access controls, retention limits, and synthetic test data
PHI sent to an unapproved model configuration	Unauthorized disclosure	Server-side gateway, provider and endpoint allowlist, configuration validation, and fail-closed behavior
Demo scope expands too far	Core safety and reliability work remains incomplete	Must-have use cases and non-goals govern backlog decisions
Clinical rules appear more complete than they are	False assurance about safety	Document supported constraints and limitations; do not market the verifier as clinical validation

Dependencies

- A working fork of the provided OpenEMR base with representative synthetic inpatient records.
- Documented OpenEMR authentication, authorization, patient data, and integration points from the required audit.
- An LLM endpoint and observability platform configured for the demo environment.
- A curated evaluation dataset with expected evidence references and safe outcomes.
- A deployment environment that supports secure secrets, structured logs, health checks, and load testing.

Open Decisions

Decision	Proposed default	Evidence needed
Rounding comparison point	Previous 24 hours with an option to use last review time	Hospitalist workflow feedback and available record timestamps
Patient identifier display	Name plus date of birth and location for demo records	Usability and privacy review
Supported laboratory rules	Display recorded abnormal flags and trends; do not independently derive clinical severity	Data audit and clinical reviewer input
Evidence navigation	Deep link to the matching OpenEMR record when feasible, otherwise show a source panel	OpenEMR integration audit
Conversation retention	End with session expiration or patient-context change	Security review and workflow feedback
Latency thresholds	Use proposed NFR1 values	Baseline testing in the deployed environment
Clinical reviewer	Qualified reviewer evaluates usefulness and citation correctness	Project access to an appropriate reviewer
OpenEMR role mapping	Map Hospitalist to the existing physician ACL or privilege	Architecture audit of roles, ACLs, facilities, and patient-access rules

Assumptions

- The first release serves adult inpatient hospitalist rounding preparation.
- Hospitalist is the only supported interactive clinical role in the MVP. Nursing workflows are deferred.
- The user is already authenticated in OpenEMR and selects a single patient before opening the copilot.
- The project uses synthetic records only and does not establish real-world HIPAA compliance or clinical validation.
- The hospitalist remains responsible for reviewing the chart and making every clinical decision.
- The OpenEMR audit may change data availability, integration choices, and the detailed delivery plan.

Traceability

Source requirement	PRD coverage
Authorized patient and role access	FR1, FR2, FR5, FR15, RBAC rules, NFR3, authorization evaluations
Source attribution and domain constraints	FR7 through FR11, FR16, claim verification policy, bounded constraint engine, safety evaluations
Speed and completeness tradeoff	Experience principles, FR3, FR4, FR17, FR18, speed and completeness policy, NFR1, latency metrics
PHI and HIPAA-aware design	Data requirements, PHI1 through PHI10, FR19, NFR3, NFR4, required security evidence, privacy risks
Failure modes, observability, and correlation IDs	FR12, FR14, FR17, FR20, failure-mode contract, NFR5, NFR6, evaluation plan, analytics, alerts, and release criteria
Boundary, invariant, and regression testing	Evaluation plan and release criteria
Health, readiness, contracts, load tests, and API collection	Data requirements, NFR2, NFR7, release criteria
Project artifacts and schedule	Delivery plan and release acceptance criteria

Product Decision Log

This PRD is the system of record for product and architecture decisions. Each accepted decision must update the affected requirements, assumptions, metrics, risks, evaluation cases, and release criteria in the same revision.

ID	Decision	Status	PRD effect
D001	The primary MVP user is a hospitalist preparing to round on admitted adult patients.	Accepted	Defines the workflow, data priorities, experience, and success measures.
D002	The MVP provides an on-demand rounding brief and patient-scoped follow-up questions inside OpenEMR.	Accepted	Defers proactive monitoring, autonomous alerts, and cross-patient summaries.
D003	A valid BAA is assumed for the selected model provider, so minimum-necessary PHI may be processed by the approved model configuration.	Accepted	Adds PHI1 through PHI10 and preserves authorization, verification, retention, and audit controls.
D004	Hospitalist is the only supported interactive clinical role in the MVP. Nursing and other clinical roles are deferred.	Accepted	Adds FR15, default-deny RBAC, role tests, and OpenEMR role mapping.
D005	Every accepted decision must be incorporated into the PRD.	Accepted	Makes this decision log and the affected PRD sections part of each revision.
D006	The product remains a tightly scoped MVP. New scope is accepted only when it supports the core workflow, a hard gate, or a critical control.	Accepted	Adds explicit MVP boundaries and defers optional roles, workflows, integrations, automation, and platform abstractions.
D007	Use one deterministic clinical constraint engine with a bounded demo rule set for allergy conflicts, laboratory ranges, medication status, two or three doses, and two or three interaction pairs.	Accepted	Adds FR16, rule contracts, constraint outcomes, safety boundaries, and focused evaluations.
D008	Use parallel prioritized retrieval, a fixed response deadline, and explicitly labeled verified partial responses.	Accepted	Adds FR17, FR18, response statuses, source-completion measures, and timeout evaluations.
D009	Default the initial rounding brief to the previous 24 hours and retrieve older detail through follow-up questions.	Accepted	Bounds initial retrieval and requires the comparison window to be displayed.
D010	Do not stream unverified model output to the clinician.	Accepted	Requires generation and verification before any clinical claim is displayed.
D011	The build must generate auditable documentation, configuration evidence, and security tests that demonstrate the PHI and authorization controls.	Accepted	Adds FR19, required evidence artifacts, traceability, and release acceptance criteria.
D012	Use exactly five primary success metrics. Track incorrect completeness as a diagnostic measure and enforce it as a zero-tolerance release criterion rather than a standalone executive metric.	Accepted	Simplifies the hospital CTO scorecard while preserving completeness safety.

ID	Decision	Status	PRD effect
D013	Define failure handling as an explicit MVP contract with four response states, no silent failure, transparent user status, and correlated structured operational events.	Accepted	Adds FR20, the failure-mode matrix, observability fields, evaluations, traceability, and release acceptance criteria.

Source and Document Control

This PRD was derived from the user-supplied Week 1 AgentForge project requirements and the selected hospitalist workflow. Proposed product targets, defaults, and design choices are labeled as such. The OpenEMR audit, user feedback, and deployed-system measurements should replace assumptions as evidence becomes available.

Version	Date	Change
0.8	15 September 2026	Added the explicit failure-mode contract, no-silent-failure rule, structured failure telemetry, and release gate
0.7	15 September 2026	Reduced the primary scorecard to five metrics and retained completeness accuracy as a release gate
0.6	15 September 2026	Required security documentation, configuration evidence, and traceable tests during buildout
0.5	15 September 2026	Added bounded clinical constraints and the fixed-time verified-partial-response policy
0.4	15 September 2026	Added the tight-MVP governance rule and explicit implementation boundaries
0.3	15 September 2026	Added BAA-based PHI processing requirements and the product decision log
0.2	15 September 2026	Added hospitalist-only MVP role model, nurse exclusion, authorization matrix, and OpenEMR role mapping
0.1	15 September 2026	Initial hospitalist-focused product requirements draft